Information Systems and Data Modeling
IT1090

Group Assignment

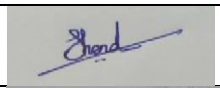
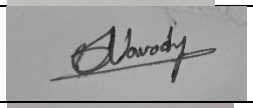
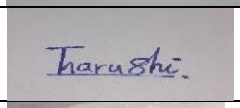
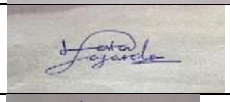
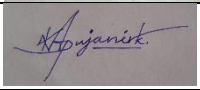
Title: Vehicle Rental System	
Batch Number: 04.02	Group Number: 02

Declaration:

We hold a copy of this assignment that we can produce if the original is lost or damaged.

We hereby certify that no part of this assignment has been copied from any other group's work or from any other source. No part of this assignment has been written / produced for our group by another person except where such collaboration has been authorized by the subject lecturer/tutor concerned.

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Introduction

In today's fast-paced world, with the advancement of technology, it has been indispensable for every business to have an online system to survive in this world with endless competitors. This is the same with the vehicle rental system. An online website enables the vehicle rental system to not be confined by the limits, as an online system provides many features in creating a thriving business. With the development of an online website, the system has been able to cater to all the demands made by customers with a vast number of resources. The online website provides the opportunity to create a strong connection between the business owners, customers, vehicle suppliers and the drivers assigned to the service and enables them to act in sync as this website acts as a bridge between these parties

. Not only the business owners of the vehicle rental system, but also the customers, vehicle suppliers, the drivers assigned to the service and everyone who plays a part in this system has been able to flourish with the benefits provided by the online vehicle rental system. The website makes it feasible for the administrators to handle the demands of the customer and make the system up to date, which makes the system error free and seamless. It is possible to generate reports through the system which in turn immensely helps to make business decisions, accordingly, leading to the success of the business. The online vehicle rental system maximizes the customer experience with user-friendliness and provides a platform to satisfy the user's needs fast and accurately.

The vehicle owners who expect to provide their vehicles for the rental service are also being immensely benefitted since this online vehicle rental system addresses the fears and provides a safe environment for the vehicles and satisfy the vehicle owner's income expectations in an organized manner, with no room for confusions. Drivers who are employed in the business of the vehicle rental system have been provided with payment in a well-planned manner and have been given job stability with the ceaseless prosperity of the online vehicle rental system. This report demonstrates the pillars which lay the foundation for the online vehicle rental system, by mentioning all the required details about the personas of the system, user journey for each persona by individually being attentive towards each party and the wireframes which suggest the basic structure of the website.

Hypothetical Scenario

In our vehicle rental system, we have developed a scenario where the platform connects customers with rental vehicles through a seamless and efficient process. The system manages the involvement of key factors such as the vehicle provider, driver, and rental agent, who work collaboratively to deliver smooth operations.

The vehicle provider offers various vehicles for rent, ensuring that each vehicle is maintained and ready for customer use. Drivers are available upon request for customers who prefer chauffeur services, while the rental agent oversees the reservation process, confirming bookings, and handling customer queries. Branches are spread across different locations to facilitate vehicle availability and returns. Once confirmed the reservations are transformed to rentals

The system allows for real-time vehicle availability, secure payment options, and the ability to track the rental status. In addition to that, system also securely transfer the monthly salary to vehicle providers and staff of the system. The vehicle rental system is also built in a way that it provides for the dependents of its staff.

Requirement Analysis

Main Requirements

Functional Requirements

Functional requirements of the vehicle rental system are crucial to meeting the needs of users Some of the roles to be performed include vehicle search and reservation, the scheduling of rentals and payments, and user account management. The system should show the availability of vehicles. Should allow the preference of customers and the duration of rent and should offer straightforward service to customers.

The system should manage the inventory of vehicles, and track the renting of vehicles, and follow through on vehicle maintenance in progress The system should grant access to administrators only for functions Examples include reporting, customer care, and rental history tracking to enhance the operational efficiency. Clearly defining these requirements facilitates the system to provide seamless rental experiences to customers and administrators.

Vehicle Owner (can access the front-end of the system)

User Requirements

- Vehicle owner can manage his vehicles by adding, updating and removing vehicle information
- Vehicle owners can manage the bookings for their vehicles.
- Vehicle owners can manage their user profiles by updating their details and business-related information.
- They can access the history of the rentals for their vehicles, payment transactions and feedback.
- Vehicle owners have the option to report and update the condition of their vehicles.
- Vehicle owners can manage the insurance documents and get notified when they need to be updated or renewed.

System Requirements

- System validates secure login and registration process for vehicle owners with multi-factor authentication.
- System can store vehicle owner information and allow real-time updates.
- System can display the bookings where vehicle owners can view and manage rental requests.
- System can store feedback on the condition and performance of rented reported by customers and drivers.
- The system ensures data privacy and system security with encryption for sensitive information.

Driver (can access the front-end of the system)

User requirements:

- Driver can check the availability of bookings and confirm the bookings.
- Driver can update the status of the trip.
- Driver can update vehicle mileage after each trip.
- Driver can report the problems.
- Driver can view customer feedback and respond to feedback.

System requirements

- System should send notifications to drivers for booking updates or changes.
- System should check the bookings records of driver.
- System should record the deposit payments details.
- System should record vehicle mileage updates after each trip.
- System should check vehicle maintenance history and display any upcoming service requirements to the driver.
- System should record the date and time of the deposit made by the driver.

Staff Functional Requirements (can access the front end of the system)

User Requirements:

- Staff can check the availability of bookings.
- Staff can report problems encountered during rentals (e.g., vehicle issues, customer complaints).
- Staff can process rental payments from customers.
- Staff can assign vehicles to bookings and update vehicle conditions.
- Staff can view their salary details, including payment history and upcoming payments.

System Requirements:

- The system should let staff easily view and manage bookings, including searching and filtering options.
- The system should record problems reported by staff and send them to the right department to fix.
- The system should securely save all payment details, including deposits and refunds.
- The system should show real-time information about vehicle availability and maintenance status.
- The system should create reports on bookings, payments, vehicle use, and staff performance.

Customer (can access the front-end of the system)

User requirements

- Customer can view and update personal profile information.
- Customer can search and check the available vehicles.
- Customer can select and reserve vehicles.
- View detailed vehicle information.
- Customer can make payments thorough the system.
- View payments history.
- Customer can cancel a reservation.
- Customer can give feedback.

System requirements

- System should allow customers to access their accounts and edit personal information.
- System should display the availability of vehicles when customers requested.
- System should check validity of details entered by customer at making account.
- System should display detailed vehicle information.
- System should validate the payment details entered by customer.
- System should allow customers to cancel reservations.
- System should allow customers to submit feedback.

Rental agent (can access the front-end of the system)

User requirements

- Rental agent must be able to register to the system as a rental agent.
- Rental agent must be able to view the rental assigned and related information.
- Rental agent must be able to accept and reject rental requests.
- Rental agent must be able to view the list of available vehicles.
- Rental agent must be able to get the monthly salary details.
- Rental agent must be able to update the status of the rental tasks.

System requirements

- System should display the all the rental tasks assigned and related information to the rental agent.
- System should display the incoming rental requests.
- System should display the available vehicles for the rental agent.
- System should add the rental tasks to the calendar.
- System should display the relevant monthly salary details.
- System should update the database accordingly.

Non-functional Requirements

Non-functional requirements describe the requirements that does not directly concerned with specific functionalities in the system. In other words, these requirements describe the characteristics or quality attributes in the system. Non-functional requirements describe of how the system should be functioned in different situations.

- **Speed**
  - The system must have good speed.
  - The system can access more users at same time without any failures.
- **Availability**
  - The system should be available in 24/7.
- **Security**
  - The system should have ability to prevent unauthorized access, misuse, forgery and secure user data. Also, by providing unique user ID and password, no one can access the system by using any other's user ID and password.
- **Performance**
  - Admin can add, edit, remove, update properties
  - Any number of users can be able to access the system at the same time and the response of the system regarding to the user requests will be very high.
- **Scalability**
  - The system should be able to handle a higher workload on-demand.
- **Reliability**
  - The system must have ability to detect the invalid user credentials.
- **User friendly**
  - The system should be accessible to users with low IT literacy.

Data Requirements

- **Reservation**
  - Reservation_ID
  - reservation_details
  - no_of_adults
  - no_of_children
  - pickup_date
  - return_date
  - pickup_itme
  - return_time
  - Branch_ID

- **Rental Agent**
  - Rental_Agent_ID
  - Staff_ID

- **Vehicle**
  - Vehicle_ID
  - Model
  - Year
  - Brand
  - Vehicle_No
  - Type
  - Color

- **Vehicle Owner**
  - Vehicle_owner_ID
  - Firstname
  - Lastname
  - NIC
  - Address
  - Contact
  - Email
  - Salary

- **Rental**
  - RentalID
  - PaymentID
  - PaymentStatus
  - JourneyStatus
  - ReservationID
  - AgentID

- **Dependent**
  - StaffID
  - Name
  - DOB
  - Age
  - Relationship

- **Payment**
  - PaymentID
  - Amount
  - Date
  - Method

- **Feedback**
  - ResponseID
  - Message
  - Date

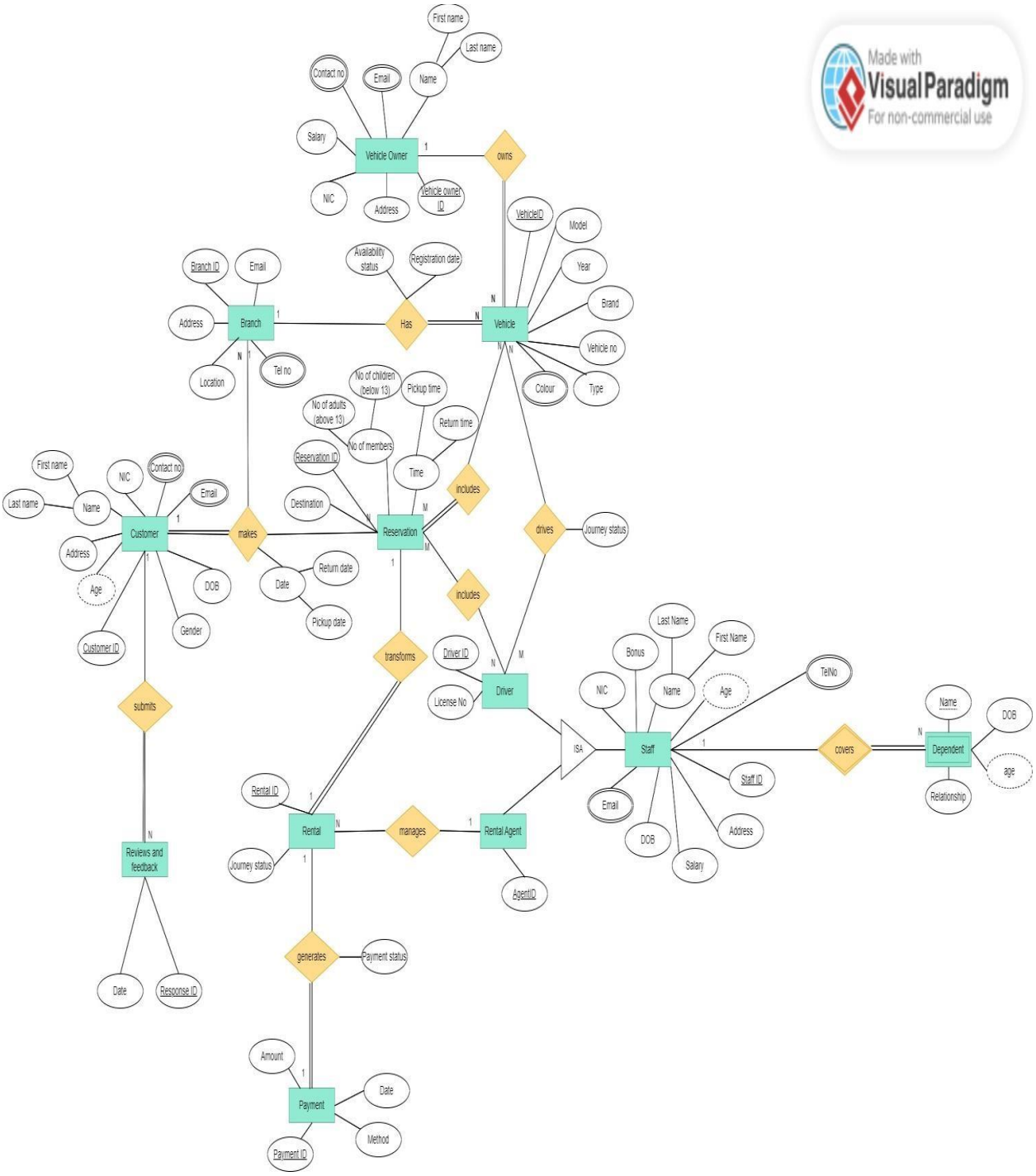
- **Staff**
  - StaffID
  - FirstName
  - LastName
  - Address
  - DOB
  - S_NIC
  - Gender
  - Salar
  - Bonus

- **Branch**
  - BranchID
  - Email
  - Location
  - Address

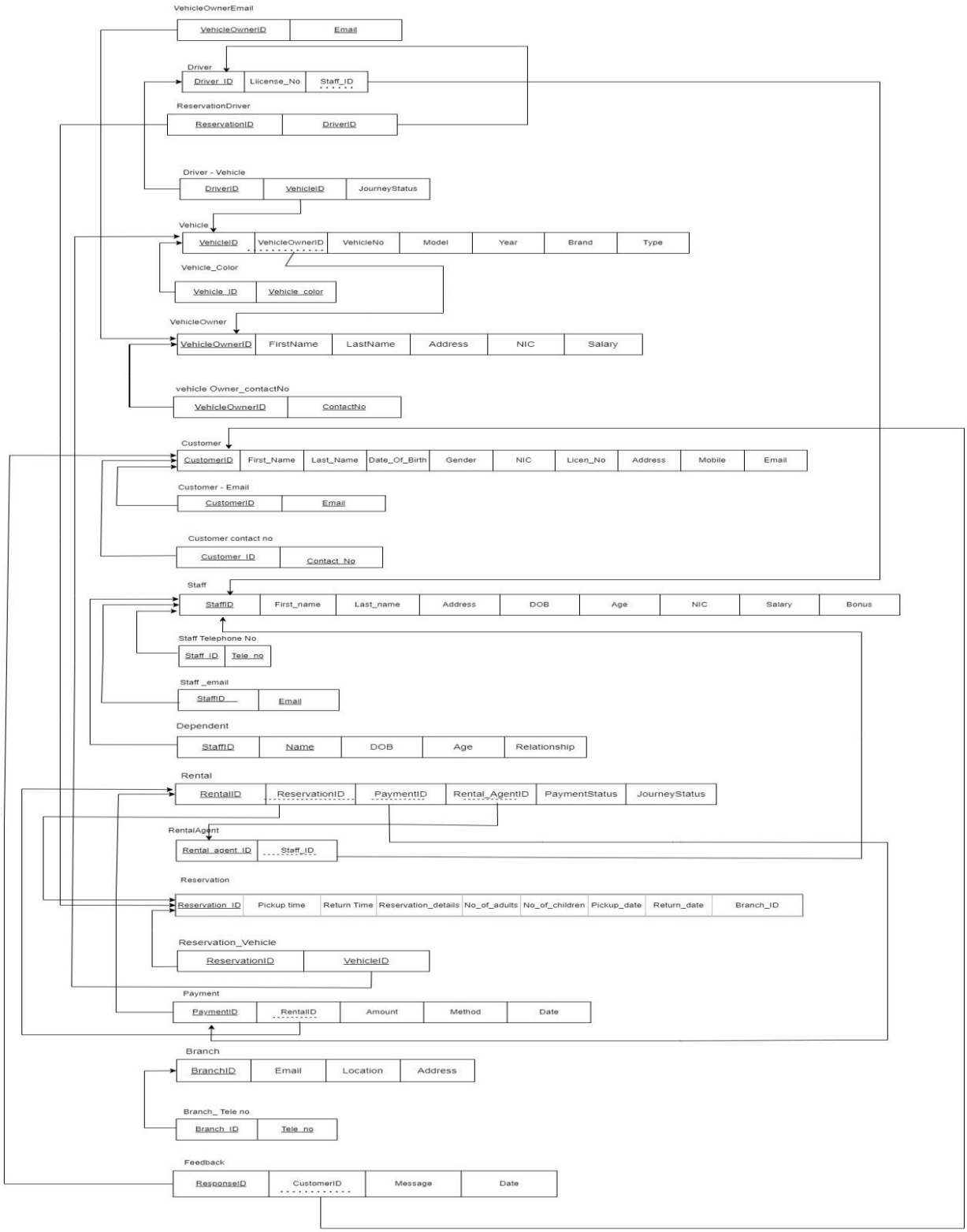
- **Customers**
  - customer_id
  - first_name
  - last_name
  - date_of_birth
  - gender
  - nic_number
  - licence_number
  - address
  - mobile
  - email

- **Drivers**
  - driver_id
  - gender
  - staff_id

ER Diagram


First name
Last name
Name
Contact no
Email
Salary
Vehicle Owner
owns
NIC
Address
Vehicle owner ID
VehicleID
Model
Year
Brand
Vehicle no
Type
Colour
Vehicle
Has
Availability status
Registration date
Branch ID
Email
Address
Branch
Location
Tel no
No of children (below 13)
Pickup time
No of adults (above 13)
Return time
No of members
Time
Reservation ID
Destination
Reservation
includes
drives
Journey status
makes
Date
Return date
Pickup date
Customer
First name
Last name
Name
NIC
Contact no
Email
Address
Age
DOB
Gender
Customer ID
submits
transforms
Driver ID
License No
Driver
includes
Last Name
Bonus
First Name
NIC
Name
Age
TelNo
ISA
Staff
Staff ID
Email
DOB
Address
Salary
covers
Name
DOB
age
Relationship
Dependent
Rental ID
Rental
manages
Rental Agent
AgentID
Journey status
Reviews and feedback
Date
Response ID
generates
Payment status
Amount
Payment
Date
Method
Payment ID

Relational Schema



SQL Queries

```
CREATE DATABASE VehicalRentalSystem;

USE VehicalRentalSystem;

--create table for vehicle owner
CREATE TABLE VehicleOwner (
    VehicleOwnerID VARCHAR(10) PRIMARY KEY,
    FirstName VARCHAR(50),
    LastName VARCHAR(50),
    Address VARCHAR(100),
    NIC VARCHAR(12),
    Salary DECIMAL(10, 2)
);
INSERT INTO VehicleOwner (VehicleOwnerID, FirstName,
LastName, Address, NIC, Salary)
VALUES
('VO101', 'John', 'Doe', '123 Elm Street, Colombo',
'823456789V', 50000.00),
('VO102', 'Jane', 'Smith', '456 Oak Avenue, Kandy',
'992345678V', 60000.00),
('VO103', 'Michael', 'Johnson', '789 Pine Road,
Galle', '200123456789', 55000.00),
('VO104', 'Sarah', 'Lee', '101 Maple Street, Jaffna',
'200234567890', 65000.00),
('VO105', 'David', 'Brown', '202 Cedar Drive,
Matara', '932345678V', 70000.00);



CREATE TABLE Customers (
    customer_id VARCHAR(5) PRIMARY KEY, -- CSXXX
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    date_of_birth DATE NOT NULL,
    gender CHAR(1) CHECK (gender IN ('M', 'F')) NOT NULL, -- 'M' for
male, 'F' for female
    nic_number VARCHAR(20) NOT NULL,
    address VARCHAR(MAX) NOT NULL, -- Use NVARCHAR(MAX) for large text
fields
    email VARCHAR(100) NOT NULL UNIQUE
);

INSERT INTO Customers (customer_id, first_name,
last_name, date_of_birth, gender, nic_number,
```

```
address, email)
VALUES
('CS001', 'John', 'Doe', '1985-01-15', 'M',
'NIC12345A', '123 Main St, City',
'john.doe@example.com'),
('CS002', 'Jane', 'Smith', '1990-05-25', 'F',
'NIC67890B', '456 Oak St, City',
'jane.smith@example.com'),
('CS003', 'Michael', 'Johnson', '1988-03-12', 'M',
'NIC23456C', '789 Pine St, City',
'michael.johnson@example.com'),
('CS004', 'Emily', 'Davis', '1992-11-05', 'F',
'NIC78901D', '321 Cedar St, City',
'emily.davis@example.com'),
('CS005', 'David', 'Brown', '1987-07-22', 'M',
'NIC34567E', '654 Birch St, City',
'david.brown@example.com');



--create table for Staff
CREATE TABLE Staff (
    staff_id CHAR(6) NOT NULL,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    address VARCHAR(255),
    age INT,
    dob DATE,
    nic CHAR(12),
    salary INT,
    bonus INT,
    CONSTRAINT staff_PK PRIMARY KEY(staff_id)
);


INSERT INTO Staff (staff_id, first_name, last_name,
address, age, dob, nic, salary, bonus) VALUES
('S0001', 'John', 'Doe', '123 Maple St, Colombo', 35,
'1989-06-15', '891234567V', 60000, 5000),
('S0002', 'Jane', 'Smith', '456 Oak Rd, Galle', 28,
'1996-09-23', '960987654V', 55000, 4000),
('S0003', 'Kamal', 'Perera', '789 Pine Ln, Kandy',
40, '1984-02-10', '841235678V', 75000, 6000),
('S0004', 'Nimal', 'Fernando', '321 Palm Ave,
Jaffna', 45, '1979-11-30', '791245678V', 80000,
7000),
```

```
('S0005', 'Amara', 'Wijesinghe', '654 Cedar Dr,
Matara', 30, '1993-03-14', '931236789V', 62000,
4500);



--create table for Branch

 CREATE TABLE Branch(
 BranchID VARCHAR(10),
 Email VARCHAR(50) NOT NULL,
 Location VARCHAR(20) NOT NULL,
 Address VARCHAR(50) NOT NULL,
 CONSTRAINT branch_pk PRIMARY KEY(BranchID)
 );

--Insert data to Branch table

INSERT INTO Branch(BranchID,Email,Location,Address)
VALUES
 ('B001', 'branch1@example.com', 'Downtown', '101
Main St, Springfield'),
    ('B002', 'branch2@example.com', 'Uptown', '202
Oak St, Springfield'),
    ('B003', 'branch3@example.com', 'Westside', '303
Pine Ave, Springfield'),
    ('B004', 'branch4@example.com', 'Eastside', '404
Maple Dr, Springfield'),
    ('B005', 'branch5@example.com', 'Southside', '505
Elm St, Springfield');



--create table for Driver
CREATE TABLE Drivers (
    driver_id VARCHAR(5) PRIMARY KEY, -- DRXXX
    licence_number VARCHAR(20) NOT NULL,
    staff_id CHAR(6),

 FOREIGN KEY (staff_id) REFERENCES Staff(staff_id) -- Define foreign
key constraint
);
```

```sql
--Insert data to driver table
INSERT INTO Drivers (driver_id, licence_number, staff_id)
VALUES
('DR001', 'LIC98765A', 'S0001'),
('DR002', 'LIC65432B', 'S0002'),
('DR003', 'LIC32109C', 'S0003'),
('DR004', 'LIC45678D', 'S0004'),
('DR005', 'LIC12345E', 'S0005');
```

```sql
--create table for vehicle
CREATE TABLE Vehicle (
    VehicleID VARCHAR(10) PRIMARY KEY,
    VehicleOwnerID VARCHAR(10),
    VehicleNo VARCHAR(15) NOT NULL,
    Model VARCHAR(50),
    Year INT,
    Brand VARCHAR(50),
    Type VARCHAR(50),
    FOREIGN KEY (VehicleOwnerID) REFERENCES
     VehicleOwner(VehicleOwnerID)
);

--Insert data to Vehicle table
INSERT INTO Vehicle (VehicleID, VehicleOwnerID,
VehicleNo, Model, Year, Brand, Type)
VALUES
('V0001', 'VO101', 'ABC-1234', 'Civic', 2020, 'Honda',
'Car'),
('V0002', 'VO102', 'XYZ-5678', 'Corolla', 2019,
'Toyota', 'Car'),
('V0003', 'VO103', 'MNO-2345', 'Ninja 300', 2021,
'Kawasaki', 'Motorcycle'),
('V0004', 'VO104', 'JKL-8765', 'Sprinter', 2018,
'Mercedes-Benz', 'Van'),
('V0005', 'VO105', 'PQR-5432', 'Hiace', 2022,
'Toyota', 'Bus');
```

```sql
--create table for vehicle_color
```

```sql
CREATE TABLE Vehicle_Color (
  VehicleID VARCHAR (10),
  Vehicle_color VARCHAR(10),
  PRIMARY KEY (VehicleID , Vehicle_color),
  FOREIGN KEY (VehicleID) REFERENCES Vehicle (VehicleID)
);

INSERT INTO Vehicle_Color (VehicleID, Vehicle_color) VALUES
('V0001', 'Red'),
('V0002', 'Blue'),
('V0003', 'Black'),
('V0004', 'White'),
('V0005', 'Silver');



--create table for VehicleOwner_ContactNo
CREATE TABLE VehicleOwner_ContactNo (
    VehicleOwnerID VARCHAR(10),
    ContactNo VARCHAR(10),
    PRIMARY KEY (VehicleOwnerID, ContactNo),
    FOREIGN KEY (VehicleOwnerID) REFERENCES
VehicleOwner(VehicleOwnerID)
);

--Insert data to VehicleOwner_contactNo table

INSERT INTO VehicleOwner_ContactNo (VehicleOwnerID, ContactNo)
VALUES
('VO101', '0712345678'),
('VO102', '0712345678'),
('VO103', '0723456789'),
('VO104', '0754567890'),
('VO105', '0715678901');



--create table for VehicleOwnerEmail--

CREATE TABLE VehicleOwnerEmail(
     VehicleOwnerID VARCHAR(10),
     Email VARCHAR(50),
     CONSTRAINT VehicleOwnerEmail_pk PRIMARY
KEY(VehicleOwnerID,Email),
```

```
     CONSTRAINT VehicleOwner_fk FOREIGN KEY(VehicleOwnerID) REFERENCES
VehicleOwner(VehicleOwnerID)
     );

-- Sample data for VehicleOwnerEmail table
INSERT INTO VehicleOwnerEmail (VehicleOwnerID, Email)
VALUES
('VO101', 'tharushi@example.com'),
('VO102', 'shenal@example.com'),
('VO103', 'kanishka@example.com'),
('VO104', 'sithmi@example.com'),
('VO105', 'visakya@example.com');



--create table for Customer-Email--

CREATE TABLE Customer_Email (
    customer_id VARCHAR(5) NOT NULL,
    email VARCHAR(100) NOT NULL,
    CONSTRAINT Customer_Email_PK PRIMARY KEY(customer_id),
    CONSTRAINT Customer_Email_FK FOREIGN KEY(customer_id) REFERENCES
Customers(customer_id)
);

INSERT INTO Customer_Email (customer_id, email) VALUES
('CS001', 'john.doe@example.com'),
('CS002', 'jane.smith@example.com'),
('CS003', 'kamal.perera@example.com'),
('CS004', 'nimal.fernando@example.com'),
('CS005', 'amara.wijesinghe@example.com');



--create table for Customer_Contact_No--

CREATE TABLE Customer_Contact_No(
     customer_id VARCHAR(5),
     contact_No VARCHAR(15),
     FOREIGN KEY (customer_id) REFERENCES Customers(customer_id) --
Define foreign key constraint
);


INSERT INTO Customer_Contact_No (customer_id, contact_No)
VALUES
```

```
('CS001', '+1234567890'),
('CS002', '+1987654321'),
('CS003', '+1122334455'),
('CS004', '+1223344556'),
('CS005', '+1445566778');


--create table for Staff_Tele_No--

CREATE TABLE Staff_Tele_No(
     staff_id CHAR(6),
     telephone_No VARCHAR(15),
     FOREIGN KEY (staff_id) REFERENCES Staff(staff_id) -- Define
foreign key constraint
);


INSERT INTO Staff_Tele_No (staff_id, telephone_No)
VALUES
('S0001', '+1777888999'),
('S0002', '+1666777888'),
('S0003', '+1555666777'),
('S0004', '+1444555666'),
('S0005', '+1333444555');



--create table for Staff_Email--
CREATE TABLE Staff_Email (
    staff_id VARCHAR(10),
    Email VARCHAR(100),
    PRIMARY KEY (staff_id, Email)
);


INSERT INTO Staff_Email (staff_id, Email)
VALUES
('S0001', 'john.doe@example.com'),
('S0002', 'jane.smith@example.com'),
('S0003', 'michael.johnson@example.com'),
('S0004', 'sarah.lee@example.com'),
('S0005', 'david.brown@example.com');



--create table for Dependent
```

```sql
CREATE TABLE Dependant(
staff_id CHAR(6),-----
Name VARCHAR(20),
DOB DATE,
Age INT,
Relationship VARCHAR(20),
CONSTRAINT Dependant_pk PRIMARY KEY(staff_id),
CONSTRAINT Dependant_fk FOREIGN KEY(staff_id) REFERENCES
Staff(staff_id)

);

INSERT INTO Dependant (staff_id, Name, DOB, Age, Relationship) VALUES
('S0001', 'Sarah Doe', '1990-05-14', 12, 'Daughter'),
('S0002', 'Michael Smith', '1980-11-23', 14, 'Son'),
('S0003', 'Emily Perera', '1993-08-12', 16, 'Daughter'),
('S0004', 'Jacob Fernando', '1987-04-06', 9, 'Son'),
('S0005', 'Alice Wijesinghe', '1996-02-18', 11, 'Daughter');



--create table for RentalAgent--

CREATE TABLE Rental_Agent (
      Rental_Agent_ID VARCHAR(10) PRIMARY KEY,
      staff_id CHAR (6) NOT NULL,
);


INSERT INTO Rental_Agent (Rental_Agent_ID, staff_id) VALUES
('RA001', 'S0001'),
('RA002', 'S0002'),
('RA003', 'S0003'),
('RA004', 'S0004'),
('RA005', 'S0005');


--create table for reservation--

CREATE TABLE Reservations (
      Reservation_ID VARCHAR(10) PRIMARY KEY,
      pickup_time TIME,
      return_time TIME,
      Reservation_details VARCHAR(255),
      no_of_adults INT,
      no_of_children INT,
```

```
      pickup_date DATE,
      return_date DATE,
      BranchID VARCHAR (10)
);


INSERT INTO Reservations (
    Reservation_ID, pickup_time, return_time, Reservation_details,
    no_of_adults, no_of_children, pickup_date, return_date, BranchID
) VALUES
('RES001', '10:00:00', '18:00:00', 'SUV rental for family trip', 2, 2,
'2024-10-15', '2024-10-20', 'B001'),
('RES002', '09:30:00', '17:30:00', 'Compact car for city use', 1, 0,
'2024-10-18', '2024-10-19', 'B002'),
('RES003', '08:00:00', '16:00:00', 'Van rental for business group', 5,
0, '2024-10-22', '2024-10-25', 'B003'),
('RES004', '13:00:00', '20:00:00', 'Luxury car for wedding', 2, 0,
'2024-10-10', '2024-10-12', 'B004'),
('RES005', '07:00:00', '19:00:00', 'Bus rental for tour group', 15,
10, '2024-10-08', '2024-10-12', 'B005');

CREATE TABLE Reservation_Vehicle(
      Reservation_ID VARCHAR(10),
      VehicleID VARCHAR(10),
      CONSTRAINT Reservation_Vehicle_pk PRIMARY
KEY(Reservation_ID,VehicleID),
      CONSTRAINT Reservation2_fk FOREIGN KEY(Reservation_ID) REFERENCES
Reservations(Reservation_ID),
      CONSTRAINT Vehicle_fk FOREIGN KEY(VehicleID) REFERENCES
Vehicle(VehicleID)
      );

INSERT INTO Reservation_Vehicle(Reservation_ID,VehicleID) VALUES
            ('RES001', 'V0001'),
            ('RES002', 'V0002'),
            ('RES003', 'V0003'),
            ('RES001', 'V0004'),
            ('RES004', 'V0005');

CREATE TABLE Rental(
      RentalID VARCHAR(10),
      ReservationID VARCHAR(10),
      Rental_agent_ID VARCHAR(10),
      PaymentStatus VARCHAR(10),
      JourneyStatus VARCHAR(15),
      CONSTRAINT Rental_pk PRIMARY KEY(RentalID),
```

```
     CONSTRAINT Reservation_fk FOREIGN KEY(ReservationID) REFERENCES
Reservations(Reservation_ID),
     CONSTRAINT agent_fk FOREIGN KEY(Rental_agent_ID) REFERENCES
Rental_Agent(Rental_Agent_ID)

     );

INSERT INTO
Rental(RentalID,ReservationID,Rental_agent_ID,PaymentStatus,JourneySta
tus) VALUES
          ('R0001', 'RES001', 'RA001', 'Paid', 'Completed'),
          ('R0002', 'RES002', 'RA002', 'Pending', 'Ongoing'),
          ('R0003', 'RES003', 'RA003', 'Paid', 'Pending'),
          ('R0004', 'RES004', 'RA004', 'Failed', 'Cancelled'),
          ('R0005', 'RES005', 'RA005', 'Paid', 'Completed');


--create table for Payment--
CREATE TABLE Payment (
    payment_id CHAR(6) NOT NULL,
    rental_id VARCHAR(10),
    amount DECIMAL(10, 2),
    method VARCHAR(10),
    payment_date DATETIME,
    CONSTRAINT payment_PK PRIMARY KEY(payment_id),
    CONSTRAINT rental_FK FOREIGN KEY(rental_id) REFERENCES
Rental(RentalID)
);



INSERT INTO Payment (payment_id, rental_id, amount, method,
payment_date) VALUES
('P0001', 'R0001', 15000.00, 'Card', '2024-09-15 10:00:00'),
('P0002', 'R0002', 12000.00, 'Cash', '2024-09-16 12:30:00'),
('P0003', 'R0003', 20000.00, 'Card', '2024-09-17 14:45:00'),
('P0004', 'R0004', 18000.00, 'Cash', '2024-09-18 09:15:00'),
('P0005', 'R0005', 16000.00, 'Card', '2024-09-19 11:00:00');



--create table for Branch_TeleNo--
CREATE TABLE Branch_TeleNo (
     BranchID VARCHAR(10),
     TeleNo VARCHAR(10),
```

```
);

INSERT INTO Branch_TeleNo (BranchID, TeleNo) VALUES
('B001', '0112345678'),
('B002', '0412456789'),
('B003', '0115678901'),
('B004', '0116789012'),
('B005', '0117890123');




--create table for feedback--

CREATE TABLE Feedback (
    response_id CHAR(6) NOT NULL,
    customer_id VARCHAR(5),
    message TEXT,
    feedback_date DATE,
    CONSTRAINT feedback_PK PRIMARY KEY(response_id),
    CONSTRAINT feedback_FK1 FOREIGN KEY(customer_id) REFERENCES
Customers(customer_id)
);


INSERT INTO Feedback (response_id, customer_id, message,
feedback_date) VALUES
('R0001', 'CS001', 'Great service and friendly staff!', '2024-09-20'),
('R0002', 'CS002', 'Vehicle was clean and well-maintained.', '2024-09-
22'),
('R0003', 'CS003', 'Had an issue with the payment system, but it was
resolved quickly.', '2024-09-25'),
('R0004', 'CS004', 'Booking process was smooth, but car pickup was
delayed.', '2024-09-28'),
('R0005', 'CS005', 'Overall, a good experience. Will use again.',
'2024-10-01');

--create table for Reservation_Driver--

CREATE TABLE Reservation_Driver(
     Reservation_ID VARCHAR(10) ,
     driver_id VARCHAR(5) ,
     Primary key(Reservation_ID ,driver_id),
     FOREIGN KEY (driver_id) REFERENCES Drivers(driver_id), -- Define
foreign key constraint
```

```
     FOREIGN KEY (Reservation_ID) REFERENCES
Reservations(Reservation_ID) -- Define foreign key constraint
);


INSERT INTO Reservation_Driver (Reservation_ID, driver_id) VALUES
('RES001', 'DR001'),
('RES002', 'DR002'),
('RES003', 'DR003'),
('RES004', 'DR004'),
('RES005', 'DR005');

--create table for Driver_Vehicle--

CREATE TABLE Driver_Vehicle (
    driver_id  VARCHAR(5) NOT NULL,
    VehicleID VARCHAR(10) NOT NULL,
    journey_status VARCHAR(20),
    CONSTRAINT Driver_vehicle_PK PRIMARY KEY(driver_id , VehicleID),
    CONSTRAINT Driver_vehicle_FK1 FOREIGN KEY(driver_id ) REFERENCES
Drivers(driver_id),
    CONSTRAINT Driver_vehicle_FK2 FOREIGN KEY(VehicleID) REFERENCES
Vehicle(VehicleID )
);


INSERT INTO Driver_Vehicle (driver_id, VehicleID, journey_status)
VALUES
('DR001', 'V0001', 'On Journey'),
('DR002', 'V0002', 'Available'),
('DR003', 'V0003', 'Completed'),
('DR004', 'V0004', 'On Journey'),
('DR005', 'V0005', 'Available');
```

Performance Requirements

For a system to be successful, performance requirements are crucial. They specify the requirements and standards that must be met in terms of the functionality, effectiveness and responsiveness of the system. The following are some essential performance requirements.

- Speed

Speed-related requirements are an essential part of performance standards, defining the maximum reaction time or processing time for various system operations.

1. The site must load, and the login process should complete within a few seconds.
2. The system must load in the shortest possible time.
3. Customer actions, such as vehicle reservation or cancellation should be processed within seconds. This guarantees that clients get responses quickly and may move forward with their rental deals without experiencing unnecessary delays.
4. The system must provide information on available vehicles in just a few seconds, Customers can choose their preferred alternative without having to wait for too long thanks to this speedy browsing process.

- Scalability

Scalability refers to a system's capacity to handle increasing demand by adding resources or expanding its capabilities.

1. To do this, one may describe the maximum number of concurrent users the system can serve or define the expected performance under certain load levels.
2. The system should be able to manage a growing number of bookings and reservations. It must handle an increasing volume of vehicle reservations while maintaining responsive performance.

- Reliability

Reliability standards, such as Mean Time Between Failures (MTBF) and system uptime, are often integral to performance requirements. These metrics ensure the system operates consistently and can quickly recover from faults or failures.

1. To reduce downtime, key infrastructure systems may have a performance requirement of 99.99% uptime.

- Accuracy

Some systems call for extremely precise calculations or processing. The maximum permissible error rate or the level of precision and accuracy necessary for a given operation may be specified by performance standards.

1. The vehicle rental system must accurately reflect real-time vehicle availability. It should provide precise and up-to-date information on available vehicles, accounting for reservations, returns, and maintenance schedules. The performance standard could specify that the system must maintain an accuracy rate of at least 99% in reflecting vehicle availability.

Security Requirements

- Implement strong authentication mechanisms to ensure that only authorized users can access the system.
- Use encryption techniques to secure sensitive information, including customer data, financial records, and transaction details.
- Ensure customer privacy by applying privacy protection practices such as data anonymization, managing consent, and adhering to data protection laws.
- Strengthen the security of the database by using secure backups, limiting access, and monitoring for vulnerabilities or breaches.
- Record all user activities, system processes, and transactions for audit purposes and to identify any unusual activity.
- Apply physical security measures to protect essential components like servers, network systems, and other critical infrastructure.
- Require strong user account passwords that include a mix of uppercase, lowercase letters, numbers, and special characters.
- Limit data access and modification rights to system administrators only**.**